```
    <dependency>
      <groupId>org.cryptomator</groupId>
      <artifactId>siv-mode</artifactId>
      <version>1.3.2</version>
    </dependency>
```

SIV mode requires two separate keys: one for the MAC and one for encryption and decryption. The specification that defines AES-SIV (https://tools.ietf.org/html/rfc5297) describes how a single key that is twice as long as normal can be split into two, with the first half becoming the MAC key and the second half the encryption key. This is demonstrated in listing 12.16 by splitting the existing 256-bit PSK key into two 128-bit keys. You could also derive the two keys from a single master key using HKDF as you learned in chapter 11. The library used in the listing provides `encrypt()` and `decrypt()` methods that take the encryption key, the MAC key, the plaintext (or ciphertext for decryption), and then any number of associated data blocks. In this example you'll pass in a header and a random IV. The SIV specification recommends that any random IV should be included as the last associated data block.

> **TIP** The `SivMode` class from the library is thread-safe and designed to be reused. If you use this library in production, you should create a single instance of this class and reuse it for all calls.

Listing 12.16 Encrypting data with AES-SIV

```
var psk = PskServer.loadPsk("changeit".toCharArray());    #A
```